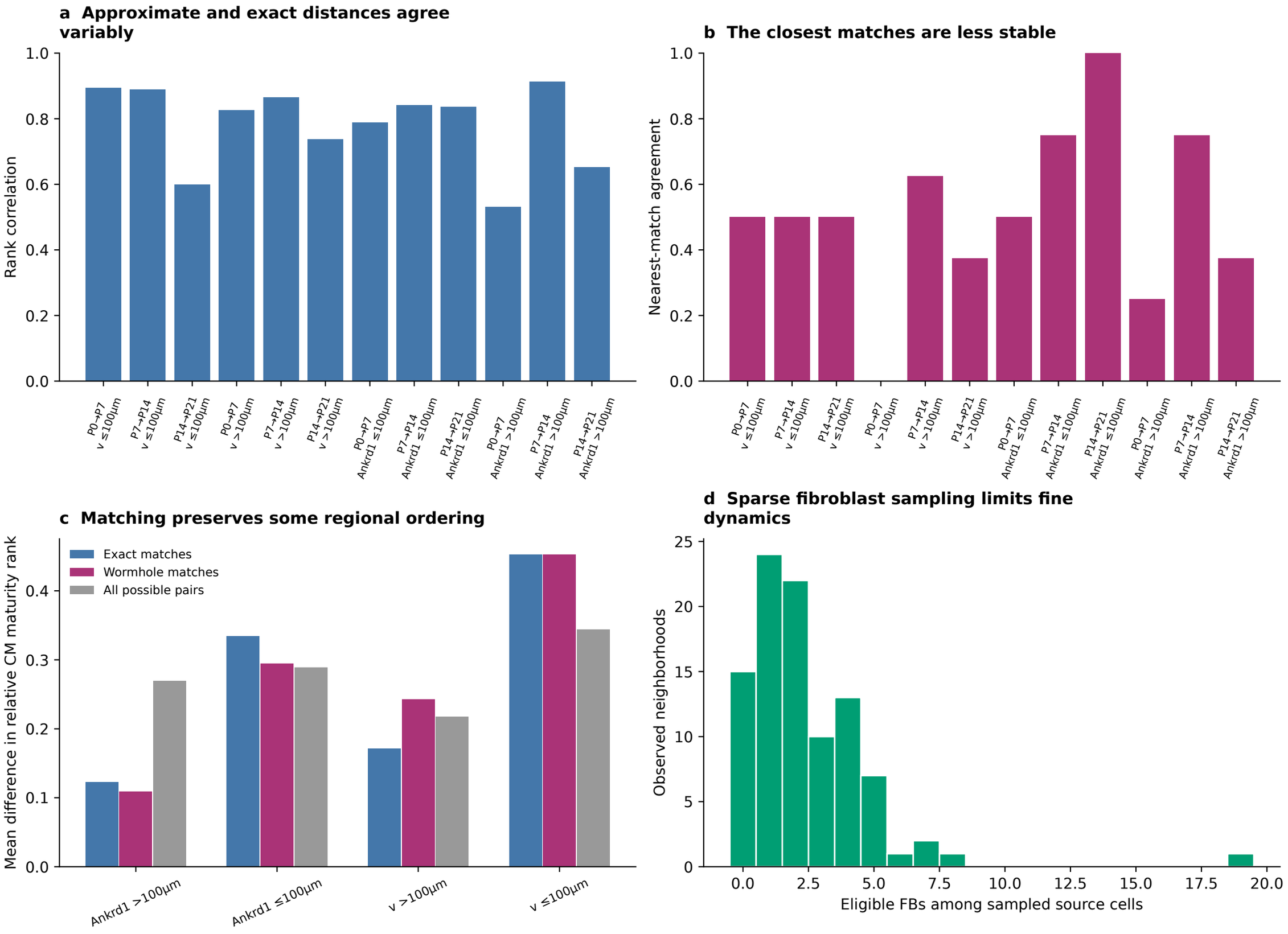
a Approximate and exact distances agree variably
Rank correlation
1.0
0.8
0.6
0.4
0.2
0.0
P0→P7 v ≤100μm
P7→P14 v ≤100μm
P14→P21 v ≤100μm
P0→P7 v >100μm
P7→P14 v >100μm
P14→P21 v >100μm
P0→P7 Ankrd1 ≤100μm
P7→P14 Ankrd1 ≤100μm
P14→P21 Ankrd1 ≤100μm
P0→P7 Ankrd1 >100μm
P7→P14 Ankrd1 >100μm
P14→P21 Ankrd1 >100μm
b The closest matches are less stable
Nearest-match agreement
1.0
0.8
0.6
0.4
0.2
0.0
P0→P7 v ≤100μm
P7→P14 v ≤100μm
P14→P21 v ≤100μm
P0→P7 v >100μm
P7→P14 v >100μm
P14→P21 v >100μm
P0→P7 Ankrd1 ≤100μm
P7→P14 Ankrd1 ≤100μm
P14→P21 Ankrd1 ≤100μm
P0→P7 Ankrd1 >100μm
P7→P14 Ankrd1 >100μm
P14→P21 Ankrd1 >100μm
c Matching preserves some regional ordering
Exact matches
Wormhole matches
All possible pairs
Mean difference in relative CM maturity rank
0.4
0.3
0.2
0.1
0.0
Ankrd1 >100μm
Ankrd1 ≤100μm
v >100μm
v ≤100μm
d Sparse fibroblast sampling limits fine dynamics
Observed neighborhoods
25
20
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Eligible FBs among sampled source cells